

C++ Assignment

1) Difference between C and C++ ?

C	C++
It is procedural oriented	It is procedural and object oriented
Function is the building block of c programming,so it is function driven	Object is the building block of C++ programming,so it is object driven
C doesn't use reference variable	C++ have use reference variable
We can't use function inside the structure is not possible	We can use the funtion inside the structure is possible
Data is not secured	Data is secured

2) How many keywords are present in C++?

Ans: Totally 63 keyword are present in C++

3) What is C++?

Ans: It is an extension and implementation of new features in C

4) What is the goal of C++?

Ans: To provide Data security

5) What are the features available in C++?

Ans:

- Encapsulation
- Data Abstraction
- Polymorphism
- Inheritance

6) what is meant by Encapsulation?

Ans: Encapsulation is the mechanism that binds together data and the member function in a single unit(class) and keeps that secure from outside interference and misuse

7) what is meant by Data Abstraction?

Ans:It refers to the act of essential features and hiding the background details

8) what is meant by Polymorphism?

Ans: A single entity can exist in more than one form. In C++ it is divided into two types

1) Compile time polymorphism

a) Function overloading

b) Operator overloading

2) Runtime polymorphism

a) Virtual function

9) What is meant by inheritance ?

Ans: It is the process of acquiring the properties of existing class (base class) into the new class (Derived class)

10) Benefits of Inheritance?

Ans: 1) code reusability and extensibility

2) Ability to add extra features to the derived class without disturbing the base class

11) what is meant by iostream ?

Ans: Combination of istream and ostream is called iostream.

C++ deals the input and output operation with streams?objects.

12) cout & cin in C++ are overloaded functions ?

Ans: Yes, cout and cin are overloaded functions.

13) cout and cin are predefined object?

Ans: cout and cin are predefined object and present inside the std namespace library.

14) What are the manipulators available in C++?

a) endl

b) hex

c) dec

d) octal

15) using namespace std ?

using --> **directive**

namespace --> **keyword**

std --> **namespace name**

16) what is meant by Namespace?

Ans:

Namespace is the keyword which is used to avoid the name collision for global variable

17) what is meant by Using namespace std?

Ans: using-----> Directive

namespace-----> keyword

std-----> namespace name

18) what is syntax of Namespace?

Ans:

namespace namespace_identifier

{

}

19) The operator >> is called____**Extraction**_____.

20) The operator << is called____**Insertion**_____.

21) __**Scope resolution operator**__ an operator, which is used to access global version of data.

22) what is meant by reference variable?

Ans: It is an alias name for existing variable which are pointing the same memory location

23) Difference between pointer vs Reference?

POINTER	REFERENCE VARIABLE
The pointer points address of the variable, which are stored in a separate memory location	It is alias name of existing variable that are stored in a same memory location
It can be initialized at any time	It should be initialized at the time of declaration
Pointer can point the NULL	It can't be NULL
Pointer can be pointed the object at any time	Reference variable can't pointed to another memory

24) Types of New Data Type in C++?

Ans: bool

string

wchar_t

25) A function declared as inline function indicates to the compiler that the body of the function can be substituted in place of the function call.

26) Encapsulation means that both the data and the methods which may access it are defined together in the same unit.

27) Size of an object is equal to size of the data member_____

28) Difference between Malloc and New?

NEW	MALLOC
New is a operator	Malloc is a function
Size of operator is not required	Size of operator is required
Memory initialization is possible	Memory initialization is not possible
Typecasting is not required	Typecasting is required

29) what is meant by Mutable in C++?

Ans: Mutable is a keyword, even though the variable is constant using mutable we can modify it.

30) what is meant by Function overloading ?

Ans: Defining multiple function with same name but arguments are different is called function overloading.

31) What is meant by Name mangling?

Ans: Name mangling is the technique used by compiler that provides name to the function based on the argument type.

32) Rules of Function Overloading in C++ ?

Ans: 1. These functions have different parameter type

2. These function have different number of parameter

3. These function have different sequence of parameter

33) what is meant by Ambiguity issue to function bind ?

Ans: In function overloading, whenever the function is called, the compiler tries to bind their exact argument matching function. At some times compiler gets confused which one is going to execute first, then at that time compiler will generate the ambiguity error.

34) what is meant by Default Arguments?

Ans: A default argument is a function parameter that has a default value is provided to it.

35) Advantages of default Arguments ?

Ans: It provides greater flexibility to the programmers

Default argument can be used to add new parameter to the existing function

It can also be used to combine similar function into one.

36) what is meant by Anonymous classes in C++ ?

Ans: Anonymous class is a class which has no name is given to it

37) difference between member function and non-member function?

Ans:

MEMBER FUNCTION	NON MEMBER FUNCTION
It is part of the class	It is not a part of the class
It is possible to access structure member directly	It is possible to access structure member with dot or arrow operator
Member function is called with dot or arrow operator	Non member function are called directly

38) what are the C++ access specifiers?

Ans: Public
Protected
Private

39) Difference between structure in c vs structure in C++?

STRUCTURE IN C	STRUCTURE IN C++
Functions are not allowed inside the structure definition	Functions are allowed inside the structure definition
Empty structure size is zero byte	Empty structure size is one byte

40) Difference between structure in c++ vs class in C++ ?

STRUCTURE IN C++	CLASS IN C++
By default public access specifier to the members	By default private access specifiers to the members
By default, public mode of inheritance	By default private mode of inheritance

41) what is meant by class?

Ans: It is an advanced structure in c++, which contains data and member function

42) why 1byte is allocated to empty struct/class in c++?

Ans: To show the object existence

In c++, if object is created at least with 1 byte, then it is possible to call the member functions

43) what is meant by Friend function?

Ans: Friend function is the non member function which is used to access the private data member of a structure or class

44) what is meant by Friend class?

Ans: Friend class is a class, which is used to access the private member of another class which it belongs as friend

45) difference between member function vs friend function ?

MEMBER FUNCTION	FRIEND FUNCTION
It is a part of the class	It is not a part of the class
This pointer concept is available	There is no this pointer
Member function should be called with object and dot operator	Friend function should be called directly
It is possible to access class member directly	It is possible to access member only with object and dot operator
The scope of member function belongs to same class	The scope of friend function can be many no of class

46) A __Constructor__ helps to perform member wise initialization.

47) A Copy Constructor takes ____One____ arguments.

48) A destructor takes ____Zero____ arguments.

49) Destructors can be overloaded (true/false) ? Ans: False

50) __Friend__ function and __Friend__ classes break the rules of the data hiding in C++. Ans: Friend function and Friend class

51) ...Friend Function... can be invoked like a normal function without the help of any object.

52) Can I create new operators using operator overloading?

a) Yes

b) NO

53) For overloaded function, the compiler chooses the right specific version on the basis of the parameter, is

A) Type B) order C) number D) all

54) The function that can not modify the object that invokes it, is said to be---

Const Member function-----

55) Why reference is not same as a pointer?

A) A reference can never be null

B) A reference once established cannot be changed.

C) Reference doesn't need an explicit dereferencing mechanism.

d) All of the above.

56) If class A is a friend of B, then B doesn't become a friend of A automatically (True / False) **Ans: True**

57) A function with the same name as the class, but preceded with a tilde character (~) is called ____**Destructor**_____ of that class.

58) How many times a constructor is called in the life-time of an object?

Ans: Only once

59) How many times a destructor is called in the life-time of an object?

Ans: Only once

60) Write the syntax for copy constructor.

Ans: Classname (Classname & ob)

```
{  
-----  
}
```

61) what is meant by Constructor ?

Ans: A constructor is a special member function of a class, which is invoked automatically when the object of the same class is created, the object gets initialized automatically by the constructor

62) Types of Constructor

Ans: 1) Default constructor

2) Parameterized constructor

3) Dynamic constructor

4) Copy constructor

63) what are the default member functions are provided by compiler in a class?

Ans: 1. Default constructor

2. Copy constructor

3. Assignment operator overload function

4. Destructor

64) what is meant by Destructor ?

Ans: A Destructor is a special member function of a class which is automatically invoked to destroy the object when the scope of the object is completed.

65) syntax of Destructor?

Ans: ~classname(void);

66) difference between shallow copy and Deep copy?

SHALLOW COPY	DEEP COPY
It is type of copy, where an object is getting initialized with another object with the help of compiler will provided an implicit copy constructor	It is type of copy, where an object is getting initialized with another object with the help of user will provided an explicit copy constructor
If an object contains data members as pointer then shallow copy will create a problem	If an objects contains data member as pointer then deep copy will not create a problem
Both objects are pointing to same memory location	Both objects are pointing to different memory location
If we modified one object then automatically another object gets affected	If we modified one object then automatically another object will not be affected

67) what is meant by this pointer?

Ans: this pointer is a keyword, which is used to store the address of the object that invokes member functions, whenever the member function is called this pointer implicitly holds the address of the object

68) difference between Assignment operator overloading vs copy constructor?

Ans:

In copy constructor, at the time of object creation object1 data is copied into object2 data.

Ex: A ob1(10),ob2;

A ob2=ob1;//internally A ob2(ob1)

In Assignment operator, first object is initialized with garbage value, then after object1 data is assigned to object2

Ex. A ob1(10),ob2;

ob2=ob1;//internally ob2.operator=(ob1)

69) What is meant by operator overloading?

Ans: The facilities of giving a special meaning to the operator without changing the existing meaning is referred to as operator overloading

70) write a syntax **Overload pre-increment operator(++)** using member function:

Ans:

```
return type operator ++(arg..)
{
-----
}
```

71) write a syntax **Overload pre-increment operator(++)** using friend function ?

Ans: return type operator ++(classname & ob)

```
{
-----
}
```

72) **Overload post-increment operator(++)** using member function ?

Ans:

```
return type operator ++(int )
{
-----
}
```

73) **Overload post-increment operator(++)** using friend function ?

Ans:

```
return type operator ++(classname & ob,int)
{
-----
}
```

74) **Overloaded operators only through member function ?**

Ans:

1. =(Assignment)
2. ()(Function call)
3. [](Subscripting)
4. -> (Arrow)

75) **Friend function cannot be overload the following operators?**

Ans:

1. =(Assignment)
2. ()(Function call)

3. [] (Subscripting)

4. -> (Arrow)

76) Why ' = ' operator can't be overloaded with friend function?

Ans:

If we define the member function for “=” (Assignment operator), then it becomes ambiguity error for compiler, when we don't define member function then compiler will supply assignment operator overload function.

77) Why [], (), -> operators can't be overloaded by friend functions?

Ans: Overloading these operators are having limitations to the language design to maintain the data security.

78) Operators that can be overloaded only by friend function?

Ans: >> ----- *Extraction operator*

<< ----- *Insertion operator*

79) Operators that cannot be overloaded by member function?

Ans: >> ----- *Extraction operator*

<< ----- *Insertion operator*

80) Q) Why '<<' & '>>' operators can't be overloaded with member functions ?

Ans: Whenever '<<' and '>>' operator are overloaded, it would require that the function be the part of the istream/ostream class, not a part of the class. since we are not allowed to modify the istream or ostream class.

81) Why cout and cin are being passed as reference objects while overloading insertion and extraction operators?

Ans: As the copy constructor and overloaded assignment operator in the classes 'istream' and 'ostream' are declared in protected session, so their objects cout and cin are passed as reference.

82) Q) What happens if 'cout' and 'cin' are passed as non-reference arguments ?

Ans: If cout and cin are passed as non-reference objects, then a new object is created on the stack or data section and tries to call the copy constructor of istream or ostream which are in protected section. As they are declared in the protected section the copy constructor cannot be called outside the class scope.

83) Operators that can't be Overloaded

Ans: 1) *.(dot)Member selection operator*

2) *.*(member selection through pointer to member operator)*

3) *sizeof operator*

4) *::(scope resolution operator)*

5) *?:(conditional operator)*

84) List of operators that can be overloaded

Ans:

operator that can be overload only by member functions

1. *=(Assignment)*

2. *()(Function call)*

3. *[](Subscripting)*

4. *->(Arrow)*

operator that can be overload only by friend function

1. *<<(Insertion)*

2. *>>(Extraction)*

85) what is meant by static data member?

Ans: 1) *If data member is declared as static means, just one copy of this data member is created for this entire class, since it is shared all the object of that class*

2) *Memory initialization is only once*

86) difference between static data member vs non-static data member>

STATIC DATA MEMBER	NON STATIC DATA MEMBER
Just one copy of the data member is created for the entire class, it can be shared all the objects of that class	Each object contains own data member which are stored in separate memory location
It is a part of the class	It is a part of the object
Static data member should not be initialized through constructor	Non-static data member should be initialized through constructor

To reduce the memory space	Increases the memory space
----------------------------	----------------------------

87) what is meant by static member function?

Ans: It is likely an ordinary function declaration, and it is a part of the class

Syntax:

```
static return type fun_name(arg)
{
    -----
}
```

88) what is meant by inheritance?

Ans: An inheritance is the process of the acquiring properties of the existing class(Base class) to the new class(Derived class).

It is a feature of a process in which new classes are created from the existing class.

89) Can I inherit the constructor and destructor of a base class?

a)yes

b) NO

90) In an inheritance hierarchy of classes

a) Constructors are invoked top-down and destructors are invoked top-down

b) Constructors are invoked bottom-up and destructors are invoked top-down

c) Constructors are invoked top-down and destructors are invoked bottom-up.

d) Constructors are invoked bottom-up and destructors are invoked bottom-up.

91) Advantages of inheritance

Ans: 1. code reusability and extensibility

2. Ability to add the extra features to the derived class without disturbing the base class.

92) what is the size of derived class and sizeof base class ?

Ans: Both are same size, until without creation of extra data members in derived class

93) Types of inheritance?

Ans:

1. Single Inheritance

2. Multi-Level Inheritance
3. Multiple Inheritance
4. Hybrid Inheritance
5. Hierarchical Inheritance
6. Multipath Inheritance

94) what is meant by Virtual Base class ?

Ans: Virtual base classes are used to avoid duplication of data.

95) what is meant by function overriding?

Ans: The mechanism of providing same function name with same argument in a base class as well as derived class is called function overriding.

96) difference between function overloading vs function overriding?

FUNCTION OVERLOADING	FUNCTION OVERRIDING
Defining multiple function with same name but argument are different is called function overloading	The mechanism of providing same function name with same arguments in a base class as well as derived class is called function overriding
An example of compile time polymorphism	An example of runtime polymorphism
Function signatures should be different	Function signature should be same
In function overloading, don't need of inheritance	In function overriding, we need inheritance concept
Overloading function are in same class	Overridden function are in different class

97) what is meant by function binding?

Ans: Resolving a function call with its proper function definition is called function binding

It is of two types: 1. compile time function binding
2. Runtime function binding

98) What is meant by Compiletime Function binding ?

Ans: Resolving a function call with its proper function definitions at compile time is called compile time function binding.

Ex. operator overloading, Function overloading

99) What is meant by Runtime Function binding ?

Ans:

Resolving a function call with its proper function definition at runtime is called runtime function binding

Ex: Virtual function

100) what is meant by Early binding?

Ans: Resolving a function call with its proper function definitions at compile time is called compile time function binding.

Ex.operator overloading,Function overloading

101) what is meant by late binding?

Ans:

Resolving a function call with its proper function definition at runtime is called runtime function binding

Ex: Virtual function

102) what is meant by virtual function?

Ans: *Virtual function are the c++ language feature to achieve the runtime polymorphism.*

Virtual function is the function which is must present in base class,which may be overrided in a derived class and which informs the compiler to perform runtime polymorphism

103) what is meant by vtable?

Ans: To implement the virtual function,c++ uses a special form of late binding is called virtual table created in compile time.

The virtual table is a lookup table of addresses of only the virtual function in the corresponding class to resolve the function with the proper function call.

104) what is meant by pure virtual function?

Ans:

Definitions are not available,only declaration available in base class is known as pure virtual function.

105) syntax of pure virtual function?

Ans: virtual return_type fun_name(arg)=0;

or

return_type virtual fun_name(arg)=0;

106) what is meant by Abstract class?

Ans: If a class contains atleast one pure virtual function is called Abstract class.

107) Rules for Virtual Functions:-

- Ans:**
- 1.Virtual function should be non-static data member
 - 2.Virtual function are accessed by using pointer to objects
 - 3.Constructor can't be virtual,but destructor should be virtual
 - 4.Virtual function should be declared in public section
 - 5.Base class pointer can holds the address of derived class object address,but reverse is not possible

108) why an object can't be created in Abstract class?

Ans:

A class having pure virtual function cannot be used to instantiate object of its own.

109) what is meant by virtual destructor?

Ans: It is destructor, to makes a proper destructor calls from derived class to base class for dynamically created object.

110) Types of Templates

Ans: Function Templates
Class Templates

111) what is meant by function Template?

Ans: It is a generic funtion

We can create a single function,which can operate an many datatypes is called function template

It allows programmer to write generic functions

Syntax:

```
template <class type>
return_type function_name( type T)
{
-----
}
```

112) difference between function overloading vs function

Templates?

FUNCTION OVERLOADING	FUNCTION TEMPLATE
Defining multiple function with same name but different argument	Defining a single function which can operate at any data types
Code duplication is acquire	No code duplication acquire
Code size is more	Code size is small

113) what is meant by class Templates?

Ans: It is generic class

We can create a single class,which can operate many datatypes,instead of create a seperate class for each type,it is called class Templates

114) A template function can be called by value as well as reference.

- a) True**
- b) false**

*115) Code that you want to monitor for exceptions must have been executed from within a _____**try**_____*

116) Which of the following operator is overloaded for object cout?

- a) <<**
- b) >>**
- c) 0**
- d) =**

117) If abstract class is inherited by derived class, then_____ .

- a) Derived class should provide definition for all the pure virtual functions
- b) Derived class also become abstract if fails to implement pure virtual functions
- c) Objects of derived class can't be created if it fails to implement pure virtual functions
- d) **All of these**

118) A virtual function that has no definition within the base class is called _____**Pure virtual function**_____

119) what is meant by Exception handling?

Ans: It is an event, which occurs at runtime and which interrupt the normal flow of program execution.

Exception which occurs at runtime due to unusual condition

To handle such exception, exception handling mechanism is used.

120) what are the three keywords in Exception handling ?

Ans:

try

catch

throw

121) What is meant by try block?

Ans:

A try block will throw the exception by using throw keyword.

Syntax:

```
try
{
    throw exception_occured;
}
```

122) what is meant by throw ?

Ans:

throw is a keyword is used to raise the exception when an error is generated.

123) what is meant by catch block?

Ans: catch block will handle the exception condition which is sent by throw

syntax:

```
try
{
throw exception;
}
catch(data_type arg)
{
-----
}
```

124) what is Generic catch block ?

Ans: Generic catch block will catch any type of data when an exception is occurs

Syntax:

```
catch(...)
{
```

}

125) Whenever unhandled exception was raised then it will call *one standard library function* is terminate function

126) cout is a

- a) operator
- b) function
- c) object
- d) macro

127) An exception is caused by

- a) a hardware problem
- b) a problem in the operating system
- c) a syntax error
- d) a run time error

```
128) #include<iostream>
using namespace std;
class A
{
    A()
    {
        cout<<"Constructor"<<endl;
    }
    ~A()
    {
        cout<<"Destructor"<<endl;
    }
};
main()
```

```
{
```

```
A obj1;
```

```
}
```

Ans: Error

129) Throwing an unhandled exception causes the standard library ____ *Terminate function* _____ to be invoked

130) Can constructors be overloaded? (True/False)

Ans: True

131) ____ *>>(Extraction)* ____ operator is used to read the data from a file?

132) which of the following object manages input from file?

a) istream

b) stream

c) ifstream

d) istream

133) A class is ____

a) Data Type

b) Abstract Type

c) User Defined Type

d) All of these options

134) write down syntax of function template?

Ans: *template<class Type>*

return_type function_name(Type T)

{

}

135) write down syntax of class template?

Ans: *template<class Type>*

class class_name

{

Type t;

Type a;

};

136) Define object of template class :

classname<datatype>object

137) what is meant by single inheritance?

Ans: Derived class is inherited by only one base class.

138) what is meant by multilevel inheritance?

Ans: Derivation of a class from another derived class.

139) what is meant by Multiple inheritance?

Ans: Derived class is inherited by more than one base class

140) what is meant by Heirarchical inheritance?

Ans: Derivation of multiple derived class from same base class

141) what is meant by Hybrid inheritance?

Ans: Derivation of a class involving more than one form of inheritance

142) what is meant by Multipath inheritance?

Ans: Derivation of a class from derived class which are derived from same base class.